```
CREATE TABLE my_clustered_table (id INT, name STRING, timestamp TIMESTAMP)
USING DELTA
CLUSTER BY (id, timestamp);
```

Figure 3.2 – Example code to create a table optimized with liquid clustering

Once the clustering is setup on your table, you want to trigger it occasionally, which you can do with the OPTIMIZE command, up next.

Optimize

The `OPTIMIZE` command will trigger clustering. This is extra important when streaming data as tables are not clustered on write. `OPTIMIZE` also compacts data files, which is vital for Delta tables with numerous small files. It merges these files into larger ones, enhancing read query speed and storage efficiency, which is especially beneficial for large datasets. Knowing when to run OPTIMIZE can be an analysis exercise. Luckily, we now have predictive optimization.

Predictive optimization

Predictive optimization is a feature that runs `VACUUM` and `OPTIMIZE` for you. Your admin can enable it in **Settings** if you have the premium version of Databricks, serverless enabled, and Unity Catalog setup. Enabling predictive optimization enables you to focus on other tasks.

Now that we've covered the basics of the Medallion architecture, Auto Loader, Delta Live Tables, and some techniques for optimizing your Delta tables, get ready to follow along in your own Databricks workspace as we work through the *Chapter 3* code by project.

Applying our learning

Before we can build models, we must first ingest the data required for each project. We've discussed how Auto Loader, schema evolution, and Delta Live Tables can format your data in storage. You'll notice that the projects below use different patterns to load data into the Bronze layer.

Technical requirements

Before we begin, please review the technical requirements needed to complete the hands-on work in this chapter:

- Databricks ML Runtime includes several pre-installed libraries useful for machine learning and data science projects. For this reason, we will be using the cluster from *Chapter 2*, as it has an ML Runtime.